

(a) Rolling forecast

72 h history / 24 h horizon

History x



Known at origin z

Predispach · PD PASA · calendar · gas



context, horizon,
recent price

x, z

Linear base

$$F_h = C_h + g_h D_h$$

Calibrator

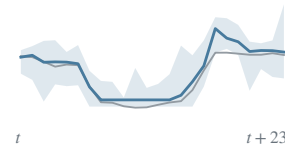
B_h

Output

inverse / sort
+ floors

δ_h

Price forecast



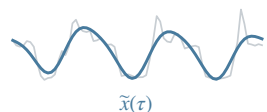
Point + 90% interval

Gray: observed price

(b) Continuous trend field

CTF / full history, not event-subtracted

Low-pass history



Fourier time features



concat



Fusion
MLP

$$c(\tau) = f_{\theta}(\tilde{x}(\tau), \gamma(\tau))$$

Trend $c(\tau)$



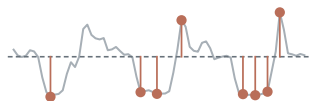
Trend head

r, z^C

(c) Detected event field

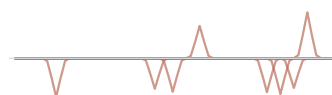
DGF / price channel only

1 Detect departures



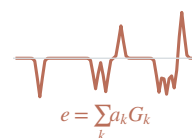
Top 8 deviations · 3 h separation
Dashed: window median

2 Form Gaussian atoms



Learned threshold κ and width w

3 Sum events



$$e = \sum_k a_k G_k$$



z^D

$$g_h = \sin^2 \theta_h$$

$g_h D_h$

Shared reconstruction: $x = c + e + r$

r also enters the trend head

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